

Transformers and Induction Machines

Topics to be covered:

- ❖ Single Phase Transformers – Principle of operation and construction of single-phase transformers, types, E.m.f equation, losses, efficiency, ratings/specifications, applications, standard codes, Illustrative problems.
- ❖ Three-phase induction motors – Principle of operation and construction, Slip-torque characteristics, types, ratings/specifications, applications, standard codes, Necessity of a starter, Illustrative examples.

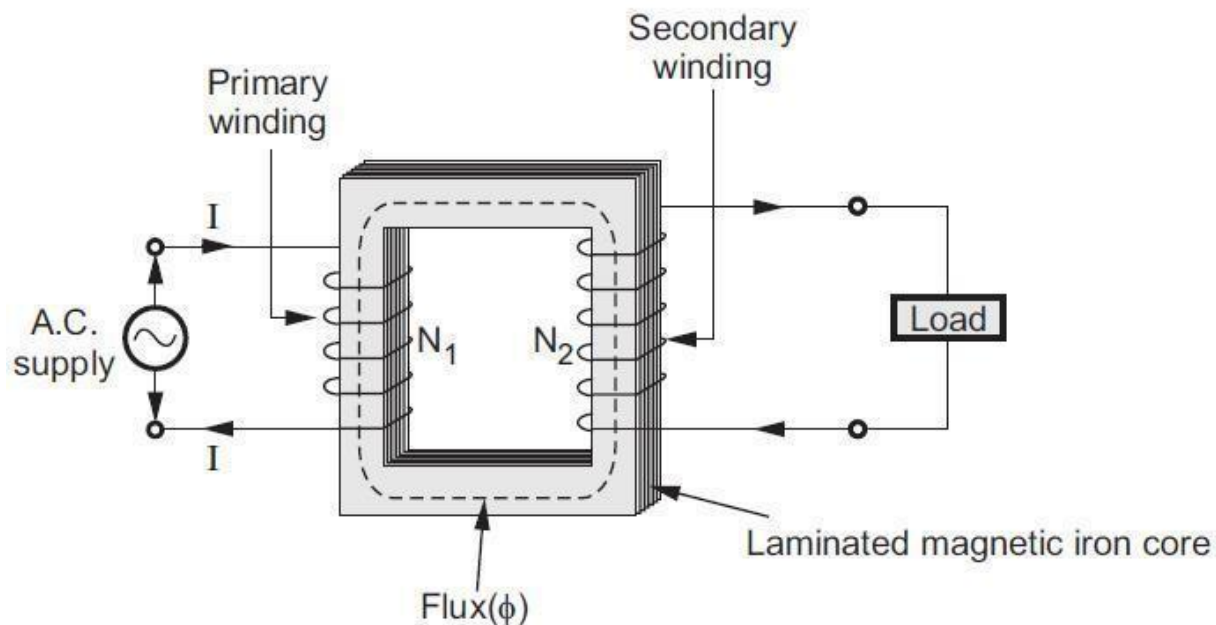
TRANSFORMER

Transformer is a static device ,which transfers electrical power from one circuit to another with required change in voltage without changing its frequency.

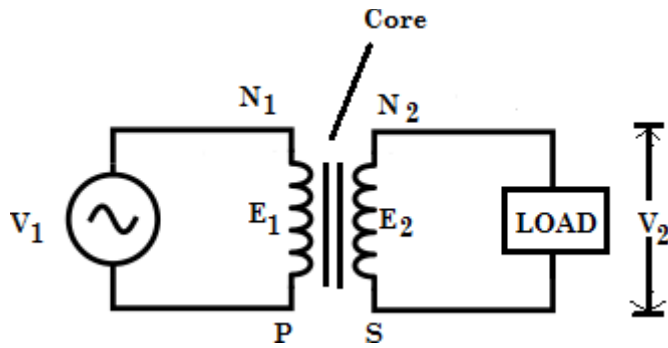
Principle of operation of single phase transformer:

Transformer works on the principle of mutual induction, which states that “when two coils are inductively coupled and if there is change in current in one of the coils, change in flux takes place in the coil which will induced an emf in the other coil wound over the same magnetic core”.

Transformer consists of two inductive coils which are electrically separated but linked through a common magnetic circuit. The two coils have high mutual inductance. The basic transformer is shown in the figure.



One of the two coils is connected to a source of alternating voltage. This coil in which electrical energy is fed with the help of source is called primary winding (P). The other winding is connected to load. The electrical energy transformed to this winding is connected to the load. This winding is called secondary winding (S). The primary winding has N_1 number of turns while the secondary winding has N_2 number of turns. When primary winding is excited by an alternating voltage, it circulates an alternating current. This current produces an alternating flux (ϕ) which completes its path through common magnetic core as shown dotted in the figure. Thus an alternating flux links with the secondary winding. As the flux is alternating, mutually induced e.m.f. gets developed in the secondary winding.



Symbolic representation of transformer

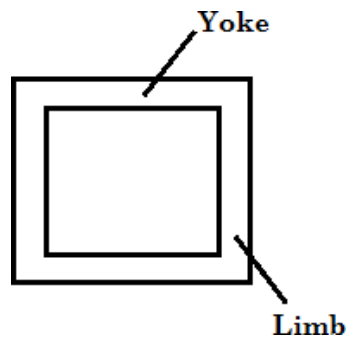
Construction of single phase transformer:

There are two basic parts of a transformer

- 1) Magnetic core
- 2) Windings or coil

1) Magnetic core

The core of the transformer is either square or rectangular in shape. It is further divided into two parts. The vertical portion on which the coils are wound is called limb, while the top and bottom horizontal portion is called yoke of the core. These parts are shown in figure



The core is made up of laminations assembled to provide a continuous magnetic path with minimum of air-gap. Generally high grade steel laminations of thickness 0.3 mm to 0.5 mm are used. The steel should have high permeability and low hysteresis loss. By effectively laminating the core, the eddy-current losses can be reduced. The laminations are insulated from each other by using insulation like paper and varnish. The transformer consists of two winding called the primary and secondary wound on a laminated iron core.

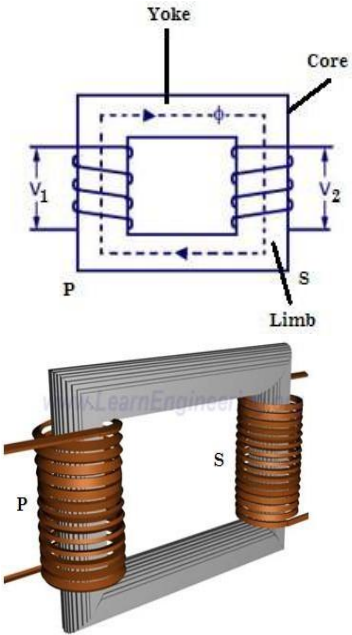
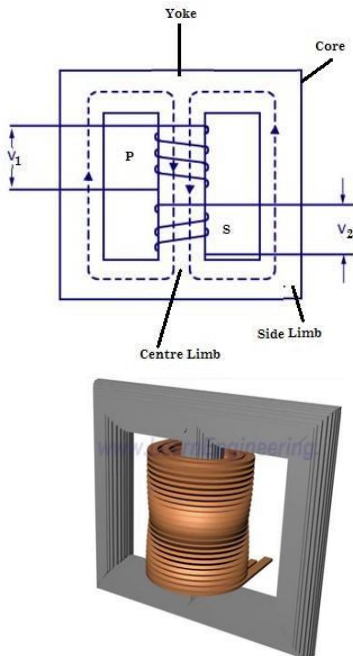
2) The windings:

The coils are wound on the limbs of the magnetic core and are insulated from each other. Primary and secondary windings are usually made up of copper. The Conductors of rectangular cross section are generally used for low voltages winding and also for the high voltage winding for large transformers. Conductors of the circular cross-sectional area are used for high voltage winding in the small transformer.

Types of Transformers:

According to the core construction and the manner in which the primary and secondary windings are wound around it, the transformer is named as core type and shell type.

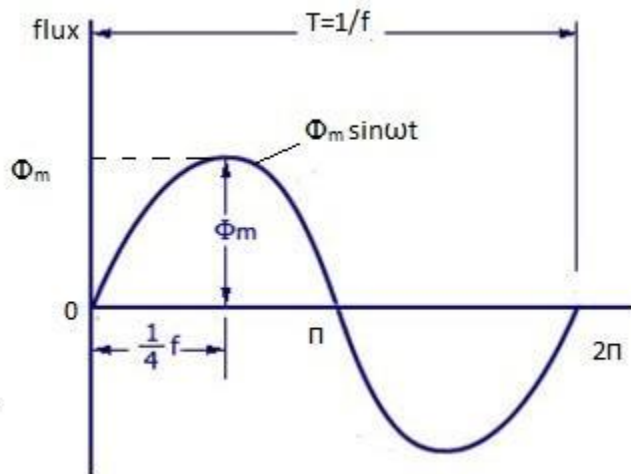
Difference between Core type and Shell type Transformer:

Sr. No.	Core Type Transformer	Shell Type Transformer
		
1)	The winding encircles the core	The core encircle most part of the winding
2)	There exists a single magnetic circuit.	There are two magnetic circuits. Total flux Φ only flows in the central limb.
3)	In Core type transformer, the core consists of two limbs and two yoke.	In Shell type transformer, the core has three limbs and two yokes.
4)	The HV and LV winding in Core type transformer are concentric.	HV and LV winding in shell type transformer is interleaved or sandwiched on the central limb.
5)	The windings are uniformly distributed on two limbs hence natural cooling is effective.	The natural cooling does not exist as the windings are surrounded by the core
6)	The coils can be easily removed for maintenance.	The coils cannot be easily removed for maintenance.
7)	Preferred for low voltage transformers.	Preferred for high voltage transformers.

E.M.F Equation of a Transformer:

When the primary winding is excited by an alternating voltage V_1 , it circulates an alternating current, producing an alternating flux Φ . The primary winding has N_1 number of turns and the alternating flux Φ linking the primary winding itself induces an emf in it denoted as E_1 . This flux links with secondary winding through the common magnetic core and induces emf E_2 in the secondary winding.

The primary winding is excited by purely sinusoidal alternating voltage. Hence the flux produced is also sinusoidal in nature having maximum value of Φ_m as shown in figure.



Let,

Φ = flux

Φ_m = Maximum value of flux

N_1 = Number of turns in primary

N_2 = Number of turns in secondary

f = frequency of the supply voltage

E_1 = R.M.S value of the primary induced emf

E_2 = R.M.S value of the secondary induced emf

From Faraday's law of Electromagnetic induction the average emf induced in each turn is proportional to the average rate of change of flux.

i.e. Average emf per turn = Average rate of change of flux.

$$\text{Average emf per turn} = d\Phi / dt$$

Now, $d\Phi/dt$ = Change in flux/Time required for change in flux

From the figure, the core flux increases from its zero value to maximum value $\Phi_{\max}$ in one quarter of the cycle, that is in $1/4$ frequency second

$$d\Phi / dt = \frac{\Phi - 0}{1/4f} = 4f\Phi_m \quad \text{Wb/second}$$

Therefore, Average emf per turn = $\Phi_m 4f$ Volts

As Φ is sinusoidal in nature, the induced emf in each turn of both the windings are also sinusoidal in nature. For sinusoidal quantity

$$\text{Form Factor} = \frac{\text{RMS value}}{\text{Average}} = 1.11$$

$$\text{value RMS value} = 1.11 \times$$

Average value

$$\text{RMS value of induced emf per turn} = 1.11 \times 4 \times f \Phi_m = 4.44 \Phi_m f$$

There are N_1 number of primary turns hence the rms value of induced emf of primary denoted as E_1 is,

$$E_1 = N_1 \times 4.44 \Phi_m f \quad \text{Volts}$$

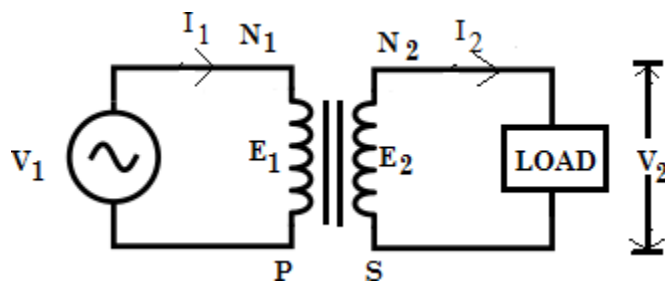
While there are N_2 number of secondary turns hence the rms value of induced emf of secondary denoted as E_2 is,

$$E_2 = N_2 \times 4.44 \Phi_m f \quad \text{Volts}$$

The expressions of E_1 and E_2 are called emf equations of a transformer.

Ratios of a Transformer:

Consider a transformer shown in figure below indicating various voltages and currents



Voltage ratio:

We know that the emf equations of a transformer are

$$E_1 = N_1 \times 4.44 \Phi_m f \quad \text{and} \quad E_2 = N_2 \times 4.44 \Phi_m f$$

Taking the ratio of two equation

$$\text{we get } \frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

This ratio of secondary induced emf to primary induced emf is known as voltage

transformation ratio denoted as K.

Thus, $E_2 = K E_1$ where $K = \frac{N_2}{N_1}$

If $N_2 > N_1$ i.e. $K > 1$, we get $E_2 > E_1$ then the transformer is called step-up transformer

If $N_2 < N_1$ i.e. $K < 1$, we get $E_2 < E_1$ then the transformer is called step-down transformer

If $N_2 = N_1$ i.e. $K = 1$, we get $E_2 = E_1$ then the transformer is called 1:1 transformer or isolation transformer

Current ratio: For an ideal transformer without any losses, the product of primary voltage V_1 and primary current I_1 , is same as the product of secondary voltage V_2 and secondary current I_2

$$V_1 I_1 = V_2 I_2$$

Therefore

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = K$$

$$V_1 I_1 = V_2 I_2$$

$$\text{Therefore, } \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{V_2}{V_1} = \frac{I_1}{I_2} = K$$

Losses in the transformer:

Two types of power losses occur in a transformer, they are

- 1) Iron loss or core loss: The core gets subjected to an alternating flux, causing core or iron loss. These are constant losses
- 2) Copper loss: the windings carry currents when transformer is loaded causing copper losses. These are varying losses

1) Iron Loss: This is the power loss that occurs in the iron part. This loss is due to the alternating frequency of the emf. Iron loss is further classified into two other losses.

- a) Eddy current loss
- b) Hysteresis loss

a) Eddy Current Loss: This power loss is due to the alternating flux linking the core, which will induced an emf in the core called the eddy emf, due to which a current called the eddy current is being circulated in the core. As there is some resistance in the core with this eddy current circulation converts into heat called the eddy current power loss. Eddy current loss is proportional to the square of the supply frequency. Eddy current loss is given by

$$\text{Eddy current loss} = k_e f^2 t^2 B_{\max}^2$$

Where, f = frequency, t = thickness, $B_{\max}$ = maximum flux density

b) Hysteresis Loss: This is the loss in the iron core, due to the magnetic reversal of the flux in the core, which results in the form of heat in the core. This loss is directly proportional to the supply frequency.

$$\text{Hysteresis loss} = k_h t f B_{\max}^{1.6}$$

Where, f =frequency, t =thickness, $B_{\max}$ =maximum flux density

Eddy current loss can be minimized by using the core made of thin sheets of silicon steel material, and each lamination is coated with varnish insulation to suppress the path of the eddy currents.

Hysteresis loss can be minimized by using the core material having high permeability.

2) Copper Loss: This is the power loss that occurs in the primary and secondary coils when the transformer is on load. This power is wasted in the form of heat due to the resistance of the coils. This loss is proportional to the square of the load current hence it is called the Variable loss whereas the Iron loss is called as the Constant loss as the supply voltage and frequency are constants. Copper losses = $I^2 R$ losses.

Efficiency of a transformer:

Due to the losses in the transformer, the output power of a transformer is less than the input power supplied.

Power output = Power input – Total losses

Power input = Power output + Total losses

$$\text{Power output} + P_i + P_{cu}$$

The efficiency of any device is defined as the ratio of the power output to power input. So for a transformer the efficiency can be expressed as

$$\eta = \frac{\text{Power output}}{\text{Power input}}$$

$$\eta = \frac{\text{Power output}}{\text{Power output} + P_i + P_{cu}}$$

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Now,

Power output = $V_2 I_2 \cos \phi$ Where,

$\cos \phi$ = Load power factor

The transformer supplies full load of current I_2 with terminal voltage V_2 P_{cu} = copper loss on full load = $I_2^2 R_{2e}$

$$\text{Therefore, } \eta = \frac{V_2 I_2 \cos \phi}{V_2 I_2 \cos \phi + P_i + I_2^2 R_{2e}}$$

But, $V_2 I_2$ = VA rating of a

transformer Therefore, η

$$= \frac{(\text{VA rating}) \cos \phi}{(\text{VA rating}) \cos \phi + P_i + I_2^2 R_{2e}}$$

$$\% \eta_{F.L} = \frac{(\text{VA rating}) \cos \phi}{(\text{VA rating}) \cos \phi + P_i + I_2^2 R_{2e}} \times 100 \quad \dots \text{Full load efficiency}$$

$$\% \eta_{F.L} = \frac{(\text{VA rating}) \cos \phi}{(\text{VA rating}) \cos \phi + P_i + (P_{cu})_{FL}} \times 100 \quad \dots \text{Full load efficiency}$$

$$\% \eta = \frac{n (\text{VA rating}) \cos \phi}{n (\text{VA rating}) \cos \phi + P_i + n^2 (P_{cu})_{FL}} \times 100 \quad \dots \text{Fractional load efficiency}$$

where, n = fractional load

For example if transformer is subjected to half

load then $n = \frac{\text{half load}}{\text{full load}} = \frac{1}{2} = 0.5$

full load $n = 1$

Condition for Maximum efficiency

To obtain the condition for maximum

$$\frac{d\eta}{dI_2} = 0$$

Condition to achieve maximum efficiency is

Copper Losses = Iron losses

i.e. $P_{cu} = P_i$

Features of Power transformer and Distribution transformers

Criteria	Power transformer	Distribution transformers
Definition	Stepping up the voltage for efficient transmission	Stepping down the voltage for consumption
Specifications	Higher voltages and commonly rated above 200 MVA	Middle to lower voltage ranges and commonly rated below 200 MVA
Efficiency	Operates at maximum efficiency (100%)	Operates around (60%-70%) efficiency
Size	Larger and Heavier	Smaller compared to power transformers
Losses	More constant, optimal losses at full load operation	More Fluctuating, optimal losses when operated around 75% of full load
Winding Connections	Usually Primary winding is connected in star, secondary in delta	Usually Primary winding is connected in delta, secondary in star.

Standards and Ratings of transformer:

The manufacturers and users in India have to follow IS(Indian Standard) and IEC(International Electrotechnical Commission) standard.

IEC60076-1 :1993 Power transformers - Part 1: General

IS 2026 (Part 1) : 1977 Power transformers : Part 1 General (first revision)

The ratings of the power transformer are as follows: 400 KV, 200 KV, 110 KV, 66 KV and 33 KV. They are mainly rated above 200 MVA.

IS 1180 (Part 1) : 1989 Distribution transformers : Part 1 General (first revision)

The ratings of the distribution transformer are as follows: 11 KV, 6.6 KV, 3.3 KV, 440 V and 230 V. They are rated less than 200 MVA.

Specifications of transformer:

Indian Standard IS 2026 “Specification for power transformer” is the governing standard on power transformers. This has been revised time to time and is now in five parts and is an exhaustive standard on power transformers.

- 1 Type of transformer
- 2 Number of phase: Single or Polyphase
- 3 Frequency
- 4 kVA rating
5. Rated voltage for each winding
6. Connection symbol
7. Impedance voltage rated current

Applications of transformer:

1. Step up or step down the level of voltage in power transmission system like transmission and distribution systems.
- 2 To isolate the low voltage circuit from high voltage circuits in case of substations, control circuitry circuits in industries.
3. Instrument transformers like current and potential transformers are used in protection and meter indication systems
4. Transformers are also used for impedance matching.

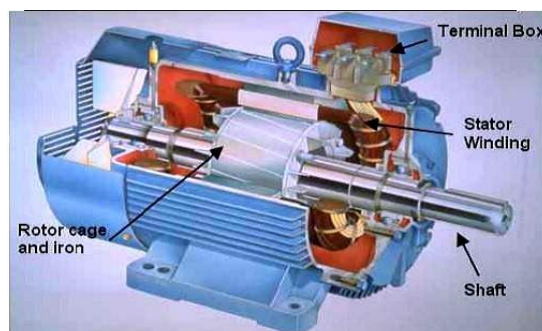
Three phase Induction Motors

The asynchronous motors or the induction motors are most widely used ac motors in industry. They convert electrical energy to mechanical energy. They work on the principle of electromagnetic induction. They are simple and rugged in construction, economical with good operating characteristics and efficiency, requiring minimum maintenance, but have a low starting torque. They run at practically constant speed from no load to full load condition. The 3 - phase induction motors are self-starting while the single phase motors are not self- starting as they produce equal and opposite torques (zero resultant torque) making the rotor stationary. The speed of the squirrel cage induction motor cannot be varied easily. The working principle of three phase induction motor is based on rotating magnetic field. Hence it is necessary to understand how the rotating magnetic field is produce.

Rotating Magnetic Field (R.M.F.)

The rotating magnetic field can be defined as the field or flux having constant amplitude but whose axis is continuously rotating in a plane with certain speed. So if the arrangement is made to rotate a permanent magnet, then the resulting field is a rotating magnetic field. But in this method, it is necessary to rotate a magnet physically to produce rotating magnetic field.

But in three phase induction motors such a rotating magnetic field is produced by supplying currents to a set of stationary windings, with the help of three-phase A.C supply. The current carrying windings produce the magnetic field or flux. Due to the interaction of three fluxes produced by three phase supply, resultant flux also has a constant magnitude and its axis rotating in space without physically rotating the windings. This type of field is nothing but rotating magnetic field.



Cut view of a three phase squirrel cage type induction motor

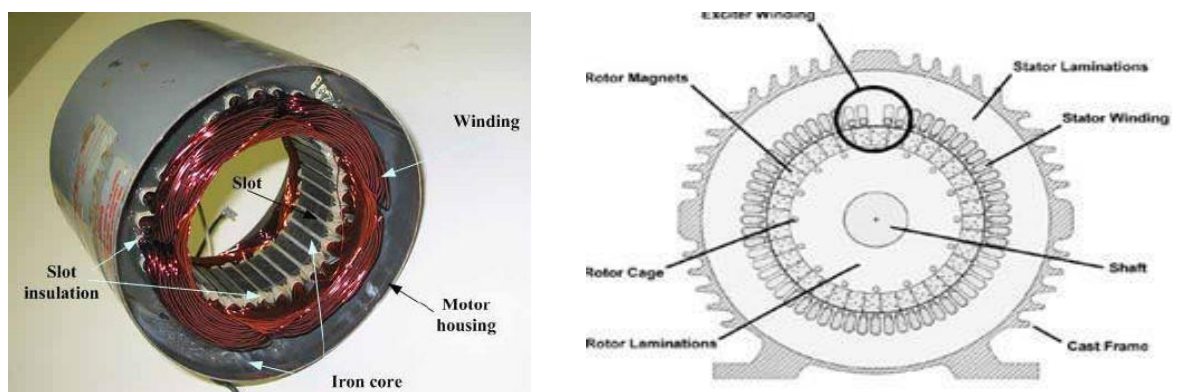
Constructional Features:

Induction motors consists of two parts, namely

1. The part which is stationary and accommodates three phase windings is called stator
2. The part which rotates and is connected to the load through shaft is called rotor



Stator: Stator is a stationary part. This consists of a core and slots to hold the stator winding to which A.C voltage is supplied. The stator is built using a cast steel frame, within this frame, laminations of silicon steel having slots in the inner periphery are stacked to form the stator core. Each lamination is insulated from each other with varnish or paper. The laminated construction is basically used to reduce the eddy current losses. The slots on the periphery of the stator core carry the three phase windings, connected in either star or delta. This three phase winding is called stator winding. It is wound for definite number of poles. This winding when excited by three phase supply produces rotating magnetic field. The choice of number of poles depends on the speed of rotating magnetic field required.



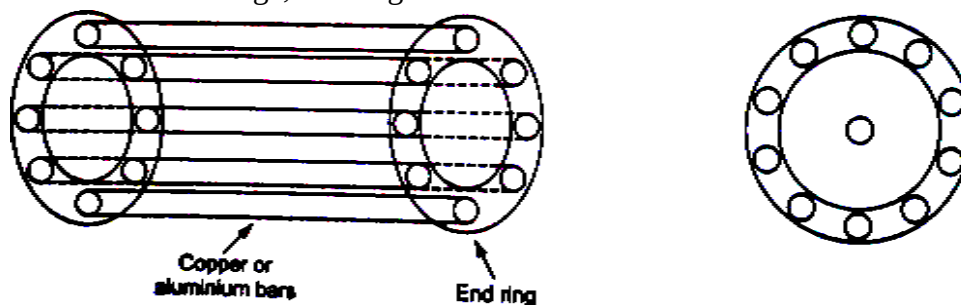
Rotor:

The rotor is placed inside the stator. The rotor core is also laminated and made of cast iron. It is cylindrical, with slots on its periphery. The rotor conductors or windings are placed in the rotor slots. The two types of rotor construction are

1. Squirrel cage rotor
2. Slip ring or wound rotor

1. Squirrel cage rotor: The rotor core is cylindrical and slotted on its periphery.

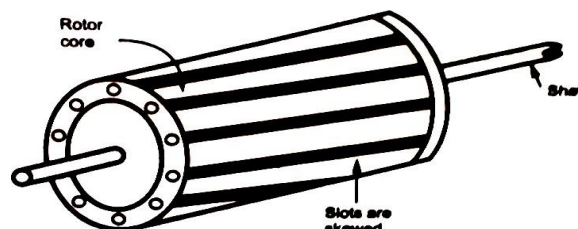
The motor consists of un-insulated copper or aluminium bars called rotor conductors. The bars are placed in the slots. The bars are permanently shorted at each end with the help of conducting copper ring called end ring. The bars are usually brazed to the end rings to provide good mechanical strength. The entire structure looks like a cage, forming a closed electrical circuit. So the rotor is called



squirrel cage rotor. The construction is as shown in figure below.

As the bars are shorted at each end with end rings, the entire rotor resistance is small. Hence this rotor is also called short circuited rotor. As rotor itself is short circuited, no external resistance can be introduced in rotor circuit. So slip ring and brush assembly is not required for this rotor. Hence construction is simple. Fan blades are provided at the ends of the rotor core. This circulates the air through the machine while operation, providing the necessary cooling. The air gap between the stator and rotor is kept uniform and as small as possible.

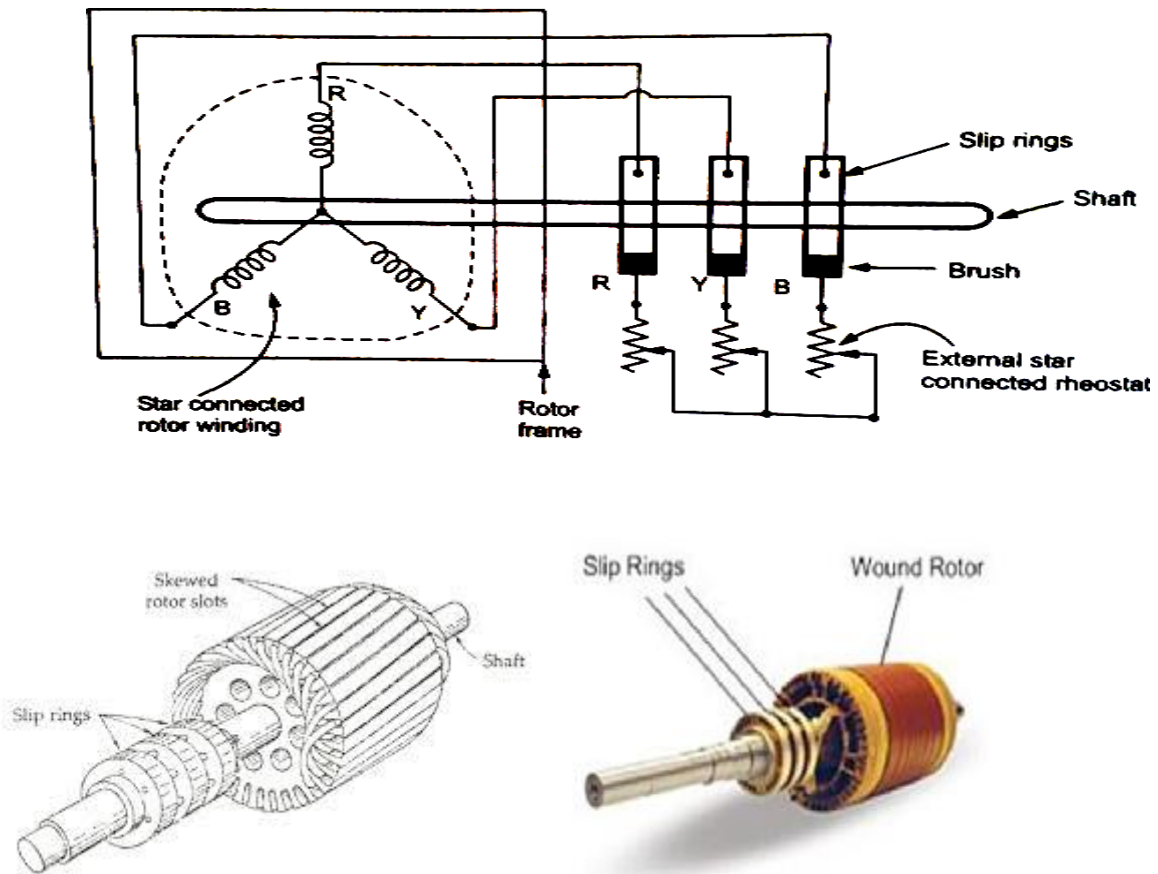
In this type of rotor, the slots are not arranged parallel to shaft axis but are skewed as shown in figure.



Skewing reduces humming noise and makes the motor operation quieter, smooth and also reduces magnetic locking of stator and rotor tooth.

2. Slip ring rotor or Phase wound rotor:

In this type of construction the rotor winding is exactly similar to stator. The rotor carries a three- phase star or delta connected distributed winding, wound for the same number of poles as that of stator. The rotor construction is laminated and slotted. The slots contain the rotor windings. The three end of the three phase winding, available after connecting the winding in star or delta, are permanently connected to the slip rings mounted on the shaft. Slip rings are used to connect external stationary circuit to the internal rotating circuit. So in this type of rotor, the external resistance can be added with the help of brushes and slip ring arrangement, in series with each phase of the rotor winding. This arrangement is shown in figure below.



In the running condition the slip rings are shorted. This is possible by connecting a metal collar which gets pushed and connects all the slip rings together, shorting them. At the same time brushes are also lifted from slip rings. This avoids wear and tear of the brushes due to friction.

Principle of Operation:

The working principle of three phase induction motor is based on rotating magnetic field. This type of rotating magnetic field is caused by three phase currents flowing in the three phase winding provided in the stator of the motor. The speed of this rotating magnetic field is known as synchronous speed (N_s). Due to the relative motion between the rotating magnetic field and the rotor, a voltage is induced in the conductors placed in the rotor. Since the conductors on the rotor forms a closed path due to end rings (in cage type of rotor) or through external resistance via slip rings (in case of slip ring or wound rotor), a current flows in them due to the induced voltage. Because of the induced current in the rotor conductors, there will be force acting on the rotor, which causes the rotor to rotate in the same direction as the rotating magnetic field. The direction of the force acting on each conductor can be determined by Fleming's left hand rule.

It is obvious that the rotor speed cannot become equal to the synchronous speed because, in that case, there will be no relative motion between the rotating magnetic field and the rotor, resulting in no induced current in the rotor conductors, and no torque acting on the rotor. Then, the rotor tends to tetrad or stop. Therefore there will be always a slight

difference between the synchronous speed (N_s) and speed of the rotor (N_r). Since the rotor never runs at the synchronous speed, induction motors are also known as asynchronous motors. Since the torque on the rotor is caused due to the induced current, these types of motors are also called induction motors.

Slip and its significance:

Slip speed of induction motor is the relative speed of the rotor with respect to the rotating magnetic field. This is given by

$$\text{Slip speed} = N_s - N_r$$

Where, N_s = synchronous speed

N_r = actual speed of the rotor or motor

Slip of the induction motor is defined as the difference between the synchronous speed (N_s) and actual speed of the rotor or motor (N_r) expressed as a fraction of the synchronous speed (N_s).

Slip is denoted as 's'.

$$\text{Thus, } s = \frac{N_s - N_r}{N_s}$$

$$\% s = \frac{N_s - N_r}{N_s} \times 100$$

In terms of slip the actual speed of the motor can be expressed as

$$N_r = N_s (1-s)$$

At start, motor is at rest and hence its speed $N_r = 0$, therefore, $s=1$

This is the maximum value of slip that can be achieved by the induction motor which occurs at start. While $s=0$ gives us $N_r = N_s$ which is not possible for induction motor.

So slip of induction motor should be $0 < s \leq 1$.

Effect of slip on rotor frequency:

In case of Induction motor the speed of the rotating

magnetic field is $N_s = 120f/P$ (1)

Where, f = frequency of the supply in Hz

At start when $N_r = 0$, $s=1$ and stationary rotor has maximum relative motion with respect to rotating magnetic field. Hence maximum voltage is induced in the rotor conductors at start. The frequency of this induced voltage is equal to the supply frequency.

As the motor actually rotates with speed N_r , the relative speed of the rotor with respect to rotating magnetic field decrease and becomes equal to slip speed $N_s - N_r$. Hence in the running condition the magnitude of the induced voltage also decreases so as its

frequency. If f_r is the frequency of the rotor induced e.m.f in running condition at slip speed $N_s - N_r$ then there exists a relation between $(N_s - N_r)$, f_r and poles P given by

$$N_s - N_r = 120f_r/P \text{----- (2)}$$

Dividing equation (2) by (1) we get

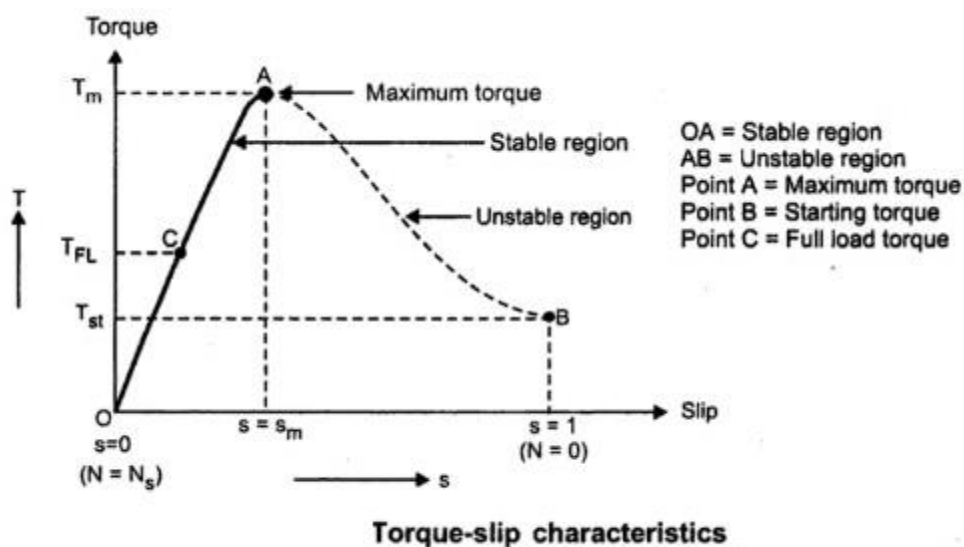
$$\frac{N_s - N_r}{N_s} = \frac{f_r}{f}$$

$$s = \frac{f_r}{f}$$

$$f_r = s f$$

Thus the frequency of the rotor induced e.m.f (f_r) in running condition is slip times the supply frequency (f).

Slip-torque characteristics:



The **Torque Slip Characteristic** is represented by a **rectangular hyperbola**

The curve obtained by plotting torque against slip from $s = 1$ (at start) to $s = 0$ (at synchronous speed) is called torque-slip characteristics of the induction motor.

The **torque slip characteristic curve** is divided into three regions.:

- Low slip region
- Medium slip region
- High slip region

Low Slip Region

At the synchronous speed, $s = 0$, therefore the torque is zero. When the speed is very near to synchronous speed. The slip is very low

Medium Slip Region

As the slip increases, the speed of the motor decreases with the increase in load.

High Slip Region

Beyond the maximum torque point, the value of torque starts decreasing. As a result, the motor slows down and stops.

The motor operates for the values of the slip between $s = 0$ and $s = s_m$. Where, s_m is the value of the slip corresponding to the maximum torque. For a typical induction motor, the pull-out torque is 2 to 3 times the rated full load torque. The starting torque is about 1.5 times the rated full load torque.

Necessity of starter for 3 phase induction motor:

When a 3- phase motor of higher rating is switched on directly from the mains it draws a starting current of about 6 -7 times the rated full load current. This heavy current may not damage the windings as it exists for short duration, but will cause a drop in the voltage affecting the performance of other loads connected to the mains. Hence starters are used to limit the initial current drawn by the 3 phase induction motors. The starting current is limited by applying reduced voltage in case of squirrel cage type induction motor and by increasing the impedance of the motor circuit in case of slip ring type induction motor

This can be achieved by using:

1. Star –delta starter
2. Auto transformer starter
3. Soft starter

Applications of Squirrel cage Induction motors:

Cage type of induction motors are now widely used in almost all industrial applications because they are rugged, almost maintenance free and economical when compared to wound rotor type induction motors. Even high starting torque can be achieved using modern deep-bar or double cage induction motors. Some applications are listed below:

- ❖ General purpose applications requiring normal torque and normal starting current – fans, blowers, centrifugal pumps etc.
- ❖ Applications requiring high torque and low starting current- conveyors, compressors, crushers, agitators, reciprocating pumps etc.

Applications of Slip ring or wound rotor Induction motors:

Wound rotor induction motors are used in situations where high starting torque is required and where the starting current should be low such as in fans, pumps, conveyors, hoists and compressors.

Standards and Ratings of three phase Induction motor:

Minimum kW rating	Upto 250 kW	250kW-300kW	300kW-750kW	750kW-1000kW
kV rating	0.440	3.3	6.6	11

Design of the motor is based on the Standards IS 325-1978.

Specifications of 3-phase Induction Motor:

The important step in selection of an induction motor for specific application is deciding the ratings considering all affecting parameters.

1. Output rating: the performed output rating for induction motors upto and including 110kW are 0.06,0.09,0.12,0.18,0.25,0.37,0.55,0.75,1.1,1.5,2.2,3.7,5.5,7.5,11,15,18.5,22,30,37,45,55,75,90,100kW.
2. Type of mounting: The mounting is to be specified like foot mounting, bed mounting etc.
3. Rated voltage and rated frequency with variations: Motors shall be able to deliver rated output
 - a. A terminal voltage differing from its rated value by not more than ± 6 percent
 - b. The frequency differing from its rated value by not more than ± 3 percent
4. The preferred voltages shall be in accordance with IS 585-1962. The voltages preferred for 3phase, 50Hz machines are 415V, 3.3kV, 6.6kV and 11kV.
5. Coordination of voltage and outputs: Recommended Rated Voltage and Output

Rated voltage(kV)	Minimum rated output(kW)
$2 < kV \leq 3.3$	100
$3.3 < kV \leq 6.6$	200
$6.6 < kV \leq 11$	1000

6. The frequency shall be standard frequency of 50Hz.
7. Class of Insulation: The class of insulation used for the winding is to be given i.e class A, E, B, F, H.
8. Ambient temperatures.
9. Type of enclosure and cooling system.
10. Method of starting and drive details.
11. Type of construction and bearing arrangements.